- increases ecosystem productivity and provide food resources
- act as water traps, filters, water flow regulators and water purifiers (forest cover and vegetation)
- climate stability (forests are essential for rainfall, temperature regulation, CO_2 absorption, which in turn regulate the density and type of vegetation)
- forest resource management and sustainable development
- maintaining balance between biotic components
- cleaning up of pollutants – microbes are the biggest degraders of molecules including many anthropogenic ones which are present in effluents, sewage, garbage and agro-chemicals
- ecological stability – the varieties and richness of species contribute to ecological stability and survival of species. Biodiverse regions are reservoirs of biological resources like food resources, gene pool, genetic resource, medicinal resources, bio-prospecting
- to provide unique aesthetic value and hot spots for Ecotourism. Along with forest resources and wildlife it has commercial significance
- an indicator of the health of the ecosystem. Endemism is a crucial indicator of richness.

11.3 Biogeographical regions of India

As per the international 'biome' type of classification based upon climate, fauna and flora and the soil conditions, India can be divided into ten different biogeographic zones, **(Fig. 11.3 and table 11.1)** namely:

Table 11.1. Biogeographical Zones Of India

S. No.	Biogeographical Zones	Biotic Provinces	Area %	Biota
1.	Trans Himalaya	Comprises mountain areas of Ladakh (J&K), North Sikkim and Lahaul and Spiti of H.P	5.7%	Alpine steppe vegetation, richest wild sheep and goat community in the world (renowned for its quality wool), Chiru, black rocked crane, etc.,
2.	Himalayas	The entire mountain chain running from western to north eastern India	7.2%	Alpine forest, orchids, rhodotendrons, wild sheep, mountain goats, shrew, snow leopard, panda etc.,
3.	Indian Desert	The extremely arid area west of the Aravalli hill range comprising of the sand desert of Rajasthan (Thar) and the salt desert of Gujarat (Kutch)	6.9%	Thorn forests, deciduous forest, wild ass (endemic), Indian bustard, camel, foxes,snakes etc.,

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